Sham Satish Thakare

Pre-Doctoral Researcher at Capital One | Gold Medal in Mathematics (2024)

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RESEARCH INTERESTS

Mathematical Machine Learning & RLVR: Token-predictive entropy confounding with completion length ($r = +0.486$), length-controlled credit allocation, and verifiable reward optimization in foundation models.
Fault-Tolerant Distributed Consensus: Dynamic quorum adaptation algorithms, latency profiling under non-uniform round-trip delays, and tail latency elimination in C++20 Raft consensus clusters (Capital One Systems Research with Yuri Shelke).
State-Matched Counterfactual Error Recovery: Difference-in-differences evaluation protocols isolating self-correction from unconditional generative capacity under frozen structural covariates.

EDUCATION & ACADEMIC HONORS

Savitribai Phule Pune University (APCOER Pune), Pune, Maharashtra, India **2020 – 2024**

Bachelor of Engineering (B.E.) in Artificial Intelligence and Data Science | **First Class with Distinction** Conferred April 2024

- **Gold Medal in Mathematics (2024):** Conferred for highest academic standing across mathematical and quantitative curricula (Calculus, Linear Algebra, Discrete Mathematics, Probability & Statistics, Optimization).
- **Academic Standing:** Cumulative GPA: **8.70 / 10.00** (170/170 Credits Completed, Total Grade Points: 1479/1700; Seat: B191102032, PRN: 72159719L).
- **Final Year Academic Rank:** Final Year SGPA: **9.53 / 10.00** (Department Rank 1).
- **Flagship Coursework Honors (Grade O / 10.0 Outstanding):** Distributed Computing (10/10), Artificial Neural Networks (10/10), Operating Systems (10/10), Cyber Security (10/10), Capstone Senior Thesis (10/10), Industry Internship (10/10).
- **Medium of Instruction:** Certified as **ENGLISH** on official university transcripts (Ref: EXAM./TRANS./BEN241203044).

RESEARCH & PROFESSIONAL EXPERIENCE

Capital One — Systems Research **06/2024 – Present**

Pre-Doctoral Researcher | Mentored by **Yuri Shelke** (Distinguished Engineer) Pune, India

- Conducted pre-doctoral systems research investigating fault-tolerant distributed state machine replication under asymmetric network partitions and non-uniform round-trip delays.
- Formulated failure-domain aware dynamic quorum adaptation algorithms and spectral latency estimators in C++20 Raft consensus clusters (quorumshift), proving non-empty quorum intersection invariants ($\mathcal{Q}_i \cap \mathcal{Q}_j eq \emptyset$).
- Eliminated p99 tail latency degradation under asymmetric partitions without violating safety or linearizability guarantees across multi-region benchmarks. Manuscript under review at *IEEE Transactions on Artificial Intelligence (IEEE TAI)*.

Independent Research — Foundation Model Reasoning & Systems **2024 – Present**

Principal Investigator / Open-Source Lead Pune, India

- Investigated token-predictive entropy confounding in reinforcement learning with verifiable rewards (GRPO), demonstrating substantial completion-length collinearity ($r = +0.486$) and developing length-controlled credit allocation estimators. Under review at *IEEE TAI 2026*.
- Formulated state-contrast continuation evaluation ($D_{\text{recovery}} = -0.1100$) in recovery_eval and stateshift to measure error recovery under frozen structural covariates. Under review at *IEEE BigData 2026* and submitted to *Elsevier AIJ*.
- Designed layered runtime boundary enforcement and deterministic state-interception architectures for autonomous tool execution safety (medirush-safeagent). In peer review at *Intelligence-Based Medicine (Elsevier)* & SSRN 7446543.

PUBLICATIONS, PREPRINTS & MANUSCRIPTS UNDER REVIEW

- **When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in Language Model Reinforcement Learning**

Sham Satish Thakare • Under Review at *IEEE Transactions on Artificial Intelligence (IEEE TAI)*, 2026. [Code] [PDF]

• **AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus**

Sham Satish Thakare, Yuri Shelke (Distinguished Engineer, Capital One) • Under Review at *IEEE Transactions on Artificial Intelligence (IEEE TAI)*, 2026 • Capital One Systems Research. [Code (C++20)]

• **recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning**

Sham Satish Thakare • Under Review at *11th IEEE Special Session on Machine Learning on Big Data (IEEE BigData 2026)*. [Code] [PDF]

• **StateShift: State-Matched Evaluation of Error Recovery in Neural Reasoning**

Sham Satish Thakare • Submitted to *Elsevier Artificial Intelligence (AIJ)*, 2026. [Code] [PDF]

• **MediRush-SafeAgent: Layered Runtime Enforcement for Healthcare-Commerce LLM Tool Execution**

Sham Satish Thakare • In Peer Review at *Intelligence-Based Medicine (Elsevier)* & SSRN Preprint 7446543. [Code] [SSRN]

TEACHING, MENTORSHIP & LEADERSHIP

• **President, Social Volunteer Club**, APCOER Pune (2023 – 2024): Elected student head of university community outreach during final undergraduate year. Directed educational mentoring in quantitative and computational skills for 200+ underprivileged high-school youth, blood donation camps, digital literacy programs, and campus sustainability drives across Pune district.

• **Teaching Assistant, Foundations of Machine Learning**, APCOER Pune (2023 – 2024): Conducted tutorial sessions, code walkthroughs, and doubt-clearing labs covering supervised learning, optimization algorithms, loss landscape geometry, and PyTorch implementations.

• **Teaching Assistant, Data Structures & Algorithms**, APCOER Pune (2023 – 2024): Led lab sessions on algorithmic complexity, graph traversal, balanced search trees, and dynamic programming. Designed automated test evaluation harnesses for student programming assignments.

TECHNICAL SKILLS & CERTIFICATIONS

• **Mathematics & Theory:** Real Analysis, Advanced Linear Algebra (Spectral Decomposition, SVD), Probability Theory & Stochastic Processes, Convex Optimization (KKT Conditions, Gradient Dynamics), Discrete Mathematics, Graph Theory.

• **Machine Learning & AI:** Reinforcement Learning (PPO, GRPO, RLVR), Transformer Architectures, Statistical Learning Theory, PyTorch, JAX, Hugging Face Transformers, NumPy, SciPy.

• **Systems Programming:** C++ (C++20 Multithreading, Lock-free Atomics), Distributed Consensus (Raft), POSIX, Linux Kernel Internals, OpenTelemetry, Git, Docker, LaTeX.

• **Standardized Exams & Credentials:** Academic IELTS Overall Band **6.5** (CEFR B2) • German APS Certificate (Prüfnummer 41183/24) • Oracle Certified Architect Associate (OCI 2024) • Oracle AI Database Foundations • Microsoft Certified: SQL AI Developer • Google AI Professional Certificate • IIT Bombay Kotlin Tutorial (Cert: 3206110F6D).